```
      *                             *******  **   *******   *******
     ***               **           **    **  **  **    **  **
    ** **            ******         **    **  **  **    **  **
   **   **   **   **  **   *****   *******  **   *******   *****
  *********   **   **  **  **   **   **        **  **        **
 **       **  **   **  **  **   **  **        **  **        **
**         **  *****   **   *****   **        **  **        *******
```

Pipe Stress Analysis and Design Program

Version: 12.06.00.48

Edition: AutoPIPE Advanced

Developed and Maintained by

BENTLEY SYSTEMS, INCORPORATED
1065 N. PACIFIC CENTER DRIVE, SUITE 450
ANAHEIM, CA 92806

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***********************************************************
**                                                       **
**              AUTOPIPE SYSTEM INFORMATION              **
**                                                       **
***********************************************************
```

SYSTEM NAME : BBOU._ASME B31.8

PROJECT ID : BBOU

PREPARED BY : ______________________________
I.O

CHECKED BY   : ______________________________
E N

1ST APPROVER : ______________________________

2ND APPROVER : ______________________________

PIPING CODE	: ASME B31.8
YEAR	: 2016
VERTICAL AXIS	: Y
AMBIENT TEMPERATURE	: 24.0 deg C
COMPONENT LIBRARY	: AUTOPIPE
MATERIAL LIBRARY	: B318-16
MODEL REVISION NUMBER	: 10

D E S C R I P T I O N

T A B L E   O F   C O N T E N T S

C O O R D I N A T E S   D A T A   L I S T I N G

```
POINT   --------COORDINATE (mm   )--------- Mass   Piping
NAME         X           Y           Z      Points Restraint
-----   ----------- ----------- ----------- ------ -----------------
*** SEGMENT A, LINE # BBOU-24-PL-012-X70-CL900_UG
A00      -369638.00     1900.00        0.00    755 Unrestrained
A01 N       8326.40     1900.00        0.00      1 Unrestrained
A01         8565.00     1900.00        0.00      0 Unrestrained
A01 F       8773.18     2016.58        0.00     10 Unrestrained
A02        13565.00     4700.00        0.00      0 Unrestrained

*** SEGMENT B, LINE # BBOU-A-100
B00        36931.20     7500.00        0.00      0 Unrestrained
B01        38231.20     7500.00        0.00      0 Unrestrained
B02        39631.20     7500.00        0.00      0 Unrestrained
B03        42131.20     7500.00        0.00      0 Unrestrained
B04        42751.20     7500.00        0.00      0 Unrestrained
B05        43351.20     7500.00        0.00      0 Unrestrained
B06        44801.20     7500.00        0.00      0 Unrestrained
B07        47751.20     7500.00        0.00      0 Unrestrained
B08        49451.20     7500.00        0.00      0 Unrestrained
B09        50651.20     7500.00        0.00      0 Unrestrained

*** SEGMENT C, LINE # BBOU-A-100_N9
B05        43351.20     7500.00        0.00      0 Unrestrained
C01        43351.20     8200.00        0.00      0 Unrestrained

*** SEGMENT G, LINE # BBOU-A-100_N4
B02        39631.20     7500.00        0.00      0 Unrestrained
G01        39631.20     8200.00        0.00      0 Unrestrained

*** SEGMENT I, LINE # BBOU-A-100_N7
B07        47751.20     7500.00        0.00      0 Unrestrained
I01        47751.20     8200.00        0.00      0 Unrestrained

*** SEGMENT O, LINE # BBOU-A-200
O00        50651.20     7500.00    -2950.00      0 Unrestrained
O01        49291.20     7500.00    -2950.00      0 Unrestrained
O02        48391.20     7500.00    -2950.00      0 Unrestrained
O03        44191.20     7500.00    -2950.00      0 Unrestrained
O04        42691.20     7500.00    -2950.00      0 Unrestrained
O05        42071.20     7500.00    -2950.00      0 Unrestrained
O06        41321.20     7500.00    -2950.00      0 Unrestrained
O07        40571.20     7500.00    -2950.00      0 Unrestrained

*** SEGMENT P, LINE # BBOU-24-PL-013-X70-CL900_AG
O07        40571.20     7500.00    -2950.00      0 Unrestrained
P01        40551.20     7500.00    -2950.00      0 Unrestrained
P01 M      39408.20     7500.00    -2950.00      0 Unrestrained
P02        38265.20     7500.00    -2950.00      0 Unrestrained
P03        36215.20     7500.00    -2950.00      0 Unrestrained
P04        33515.20     7500.00    -2950.00      0 Unrestrained
P04 M      32372.20     7500.00    -2950.00      0 Unrestrained
P05        31229.20     7500.00    -2950.00      0 Unrestrained
P06        29329.20     7500.00    -2950.00      0 Unrestrained
P07        28329.20     7500.00    -2950.00      0 Unrestrained
```

C O O R D I N A T E S   D A T A   L I S T I N G

POINT NAME	COORDINATE (mm) X	Y	Z	Mass Points	Piping Restraint
P07 M	27186.20	7500.00	-2950.00	0	Unrestrained
P08	26043.20	7500.00	-2950.00	0	Unrestrained
P09	25093.20	7500.00	-2950.00	0	Unrestrained
P10	24143.20	7500.00	-2950.00	0	Unrestrained
P10 M	23000.20	7500.00	-2950.00	0	Unrestrained
P11	21857.20	7500.00	-2950.00	0	Unrestrained
P12	20857.20	7500.00	-2950.00	0	Unrestrained
P13	19457.20	7500.00	-2950.00	0	Unrestrained
P14 N	16698.15	7500.00	-2950.00	0	Unrestrained
P14	16457.20	7500.00	-2950.00	0	Unrestrained
P14 F	16247.54	7381.26	-2950.00	0	Unrestrained
P15	11407.20	4640.00	-2950.00	0	Unrestrained
*** SEGMENT W, LINE # BBOU-24-PL-013-X70-CL900_AG_B					
P06	29329.20	7500.00	-2950.00	0	Unrestrained
W01 N	29329.20	7500.00	-4085.60	0	Unrestrained
W01	29329.20	7500.00	-5000.00	0	Unrestrained
W01 F	28414.80	7500.00	-5000.00	0	Unrestrained
W02	27629.20	7500.00	-5000.00	0	Unrestrained
W02 M	26486.20	7500.00	-5000.00	0	Unrestrained
W03	25343.20	7500.00	-5000.00	0	Unrestrained
W04	23100.20	7500.00	-5000.00	0	Unrestrained
W05 N	21771.60	7500.00	-5000.00	0	Unrestrained
W05	20857.20	7500.00	-5000.00	0	Unrestrained
W05 F	20857.20	7500.00	-4085.60	0	Unrestrained
P12	20857.20	7500.00	-2950.00	0	Unrestrained
*** SEGMENT AE, LINE # BBOU-A-300-N7					
O03	44191.20	7500.00	-2950.00	0	Unrestrained
AE01	44191.20	8200.00	-2950.00	0	Unrestrained
*** SEGMENT AZ, LINE # BBOU-24-PL-012-X70-CL900_AG					
AZ00	8565.00	1900.00	0.00	0	Unrestrained
A02	13565.00	4700.00	0.00	0	Unrestrained
AZ02N	18356.82	7383.42	0.00	0	Unrestrained
AZ02	18565.00	7500.00	0.00	0	Unrestrained
AZ02F	18803.60	7500.00	0.00	0	Unrestrained
AZ03	21565.00	7500.00	0.00	0	Unrestrained
AZ04	23965.60	7500.00	0.00	0	Unrestrained
AZ04M	25108.60	7500.00	0.00	0	Unrestrained
AZ05	26251.60	7500.00	0.00	0	Unrestrained
AZ06	26751.60	7500.00	0.00	0	Unrestrained
AZ07	31098.40	7500.00	0.00	0	Unrestrained
AZ08	34798.40	7500.00	0.00	0	Unrestrained
AZ08M	35941.40	7500.00	0.00	0	Unrestrained
AZ09	37084.40	7500.00	0.00	0	Unrestrained
B00	36931.20	7500.00	0.00	0	Unrestrained
*** SEGMENT BA, LINE # BBOU-24-PL-013-X70-CL900_UG					
P15	11407.20	4640.00	-2950.00	11	Unrestrained
BA01N	6566.86	1898.74	-2950.00	1	Unrestrained
BA01	6357.20	1780.00	-2950.00	0	Unrestrained

C O O R D I N A T E S   D A T A   L I S T I N G

POINT NAME	COORDINATE (mm) X	Y	Z	Mass Points	Piping Restraint
BA01F	6116.25	1780.00	-2950.00	749	Unrestrained
BA02	-368642.78	1780.00	-2950.02	0	Unrestrained

P I P E   D A T A   L I S T I N G

```
 Pipe ID/     Nom/ O.D.  -----Thickness(mm  )----- Spec  InsuDen/   Weight(N/m   )    ZL/ Composition/
 Material     Sch  mm    W.Th. Corr Mill Insu Ling Grav/ LingDen/  Pipe/ Ling/ Total  ZC
 CladMaterial            Clad                      InsMt CladDen   Cont  Insu/
 ---Line Class---                                        kg/m3           Clad
 ------------ ---- ----- ----- ---- ---- ---- ---- ----- --------  ----- ----- ----- ---- --------------------

 Tag No. : <None>
 24"CL900     600 609.60 15.88 3.00 1.99    0    0 0.70     0.000   2275     0  4075 1.00
 5LX-X70      NS                                            0.000   1799     0       1.00
                             0                              0.000            0
 900


 Tag No. : <None>
 28"PIPE      700 711.20 24.00 3.00 3.00    0    0 0.70     0.000   3980     0  6350 1.00
 5LX-X70      NS                                            0.000   2370     0       1.00
                             0                              0.000            0
 900


 Tag No. : <None>
 8"C03        200 219.07 18.26 3.00 2.28    0    0 0.70     0.000    885     0  1065 1.00
 5L-B         120                                           0.000    180     0       1.00
                             0                              0.000            0
 900


 Tag No. : <None>
 24"C06       600 609.60 38.89 6.00 4.86    0    0 0.70     0.000   5356     0  6880 1.00
 5LX-X60      NS                                            0.000   1524     0       1.00
                             0                              0.000            0
 900


 Tag No. : <None>
 4"C06        100 114.30 17.12 6.00 2.14    0    0 0.70     0.000    401     0   436 1.00
 5LS-B        XXS                                           0.000  34.54     0       1.00
                             0                              0.000            0
 900


 Tag No. : <None>
 6"B06        150 168.27 10.97 6.00 1.37    0    0 0.70     0.000    417     0   532 1.00
 5L-B         XS                                            0.000    115     0       1.00
                             0                              0.000            0
 150


 Tag No. : <None>
 4"B06        100 114.30 11.13 6.00 1.39    0    0 0.70     0.000    277     0   323 1.00
 5L-B         120                                           0.000  45.66     0       1.00
                             0                              0.000            0
 150
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P I P E   D A T A   L I S T I N G

```
 Pipe ID/     Nom/ O.D.  -----Thickness(mm  )----- Spec  InsuDen/   Weight(N/m   )    ZL/ Composition/
 Material     Sch  mm    W.Th. Corr Mill Insu Ling Grav/ LingDen/  Pipe/ Ling/ Total  ZC
 CladMaterial            Clad                      InsMt CladDen   Cont  Insu/
 ---Line Class---                                        kg/m3           Clad
 ------------ ---- ----- ----- ---- ---- ---- ---- ----- --------  ----- ----- ----- ---- --------------------
 Tag No. : <None>
 6"C06        150 168.27 18.26 6.00 2.28    0    0  0.70     0.000    661     0   755 1.00
 5LS-B        160                                            0.000  93.54     0       1.00
                             0                               0.000            0
 900


 Tag No. : <None>
 24C70        600 609.60 24.20    0 3.03    0    0  0.70     0.000   3419     0  5116 1.00
 5L-B         NS                                             0.000   1697     0       1.00
                             0                               0.000            0
 900
```

M A T E R I A L   D A T A   L I S T I N G

Material Name	Pipe ID	Density kg/m3	Pois. Ratio	Temper. deg C	Modulus E6 N/mm2 Axial	Hoop	Shear	Expans. mm/m	Composition
5LX-X70	24"CL900	7833.0	0.30	24.0	0.20320	0.20320	0.07815		
				60.0	0.20080			0.4154	
				12.7	0.20382			-0.1280	
				23.0	0.20327			-0.0115	
				100.0	0.19816			0.8900	
				-29.0	0.20593			-0.5971	
5LX-X70	28"PIPE	7833.0	0.30	24.0	0.20320	0.20320	0.07815		
				100.0	0.19816			0.8900	
				-29.0	0.20593			-0.5971	
				23.0	0.20327			-0.0115	
5L-B	8"C03	7833.0	0.30	24.0	0.20320	0.20320	0.07815		
				100.0	0.19816			0.8900	
				-29.0	0.20593			-0.5971	
				23.0	0.20327			-0.0115	
5LS-B	4"C06	7833.0	0.30	24.0	0.20320	0.20320	0.07815		
				100.0	0.19816			0.8900	
				-29.0	0.20593			-0.5971	
				23.0	0.20327			-0.0115	

M A T E R I A L   A L L O W A B L E   D A T A   L I S T I N G

Material Name	Pipe ID	Temper. deg C	Yield N/mm2
5LX-X70	24"CL900	24.0	482.63
5LX-X70	28"PIPE	24.0	482.63
5L-B	8"C03	24.0	241.32
5LS-B	4"C06	24.0	241.32

OPERATING TEMPERATURE AND PRESSURE DATA
STRESSES IN N/mm2

POINT NAME	CASE	PRESS. N/mm2	TEMPER deg C	EXPAN. mm/m	MODULUS E6 N/mm2	YIELD STRESS
*** SEGMENT A, LINE # BBOU-24-PL-012-X70-CL900_UG						
A00	T1	11.0000	60.00	0.415	0.20080	482.63
	T2	11.0000	12.70	-0.128	0.20382	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
A02	Same as previous point.					
*** SEGMENT B, LINE # BBOU-A-100						
B00	T1	11.0000	100	0.890	0.19816	482.63
	T2	11.0000	-29.00	-0.597	0.20593	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
B09	Same as previous point.					
*** SEGMENT C, LINE # BBOU-A-100_N9						
B05	T1	11.0000	100	0.890	0.19816	241.32
	T2	11.0000	-29.00	-0.597	0.20593	241.32
	T3	11.0000	23.00	-0.012	0.20327	241.32
C01	Same as previous point.					
*** SEGMENT G, LINE # BBOU-A-100_N4						
B02	T1	11.0000	100	0.890	0.19816	241.32
	T2	11.0000	-29.00	-0.597	0.20593	241.32
	T3	11.0000	23.00	-0.012	0.20327	241.32
G01	Same as previous point.					
*** SEGMENT I, LINE # BBOU-A-100_N7						
B07	T1	11.0000	100	0.890	0.19816	241.32
	T2	11.0000	-29.00	-0.597	0.20593	241.32
	T3	11.0000	23.00	-0.012	0.20327	241.32
I01	Same as previous point.					
*** SEGMENT O, LINE # BBOU-A-200						
O00	T1	11.0000	100	0.890	0.19816	482.63
	T2	11.0000	-29.00	-0.597	0.20593	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
O07	Same as previous point.					
*** SEGMENT P, LINE # BBOU-24-PL-013-X70-CL900_AG						
O07	T1	11.0000	60.00	0.415	0.20080	482.63
	T2	11.0000	12.70	-0.128	0.20382	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
P15	Same as previous point.					
*** SEGMENT W, LINE # BBOU-24-PL-013-X70-CL900_AG_B						
P06	T1	11.0000	60.00	0.415	0.20080	482.63
	T2	11.0000	12.70	-0.128	0.20382	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
P12	Same as previous point.					
*** SEGMENT AE, LINE # BBOU-A-300-N7						
O03	T1	11.0000	100	0.890	0.19816	241.32
	T2	11.0000	-29.00	-0.597	0.20593	241.32
	T3	11.0000	23.00	-0.012	0.20327	241.32
AE01	Same as previous point.					
*** SEGMENT AZ, LINE # BBOU-24-PL-012-X70-CL900_AG						
AZ00	T1	11.0000	60.00	0.415	0.20080	482.63
	T2	11.0000	12.70	-0.128	0.20382	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
B00	Same as previous point.					

OPERATING TEMPERATURE AND PRESSURE DATA
STRESSES IN N/mm2

POINT NAME	CASE	PRESS. N/mm2	TEMPER deg C	EXPAN. mm/m	MODULUS E6 N/mm2	YIELD STRESS
*** SEGMENT BA, LINE # BBOU-24-PL-013-X70-CL900_UG						
P15	T1	11.0000	60.00	0.415	0.20080	482.63
	T2	11.0000	12.70	-0.128	0.20382	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
BA02	Same as previous point.					

u User-defined value
* Non-code material for allowable stress;
   Non-standard material for expansion and modulus

S E G M E N T   D A T A   L I S T I N G

Segment Name	First Node	Last Node	Line Number	Apply Wind	Apply Bowing	Apply Buoyancy
A	A00	A02	BBOU-24-PL-012-X70-CL900_UG	No	No	No
B	B00	B09	BBOU-A-100	Yes	No	No
C	B05	C01	BBOU-A-100_N9	Yes	No	No
G	B02	G01	BBOU-A-100_N4	Yes	No	No
I	B07	I01	BBOU-A-100_N7	Yes	No	No
O	O00	O07	BBOU-A-200	Yes	No	No
P	O07	P15	BBOU-24-PL-013-X70-CL900_AG	Yes	No	No
W	P06	P12	BBOU-24-PL-013-X70-CL900_AG_B	Yes	No	No
AE	O03	AE01	BBOU-A-300-N7	Yes	No	No
AZ	AZ00	B00	BBOU-24-PL-012-X70-CL900_AG	Yes	No	No
BA	P15	BA02	BBOU-24-PL-013-X70-CL900_UG	Yes	No	No

SOIL DATA LISTING

SOIL STIFFNESS PROPERTIES (24SAND)

	Dirn	Auto	Initial K (N/mm/mm)	Auto	Yield P (N/m)	Final K (N/mm/mm)	Yield disp (mm)
Low Stiffness							
	Horiz.	Y	3.737	Y	227788.59	0.000	60.9600
	Long.	Y	7.654	Y	19441.434	0.000	2.5400
	Vert. Up	Y	2.176	Y	33167.953	0.000	15.2400
	Vert. Dn	Y	20.633	Y	1257770.0	0.000	60.9600
High Stiffness							
	Horiz.	Y	3.737	Y	227788.59	0.000	60.9600
	Long.	Y	7.654	Y	19441.434	0.000	2.5400
	Vert. Up	Y	2.176	Y	33167.953	0.000	15.2400
	Vert. Dn	Y	20.633	Y	1257770.0	0.000	60.9600
Average Stiffness							
	Horiz.	Y	3.737	Y	227788.59	0.000	60.9600
	Long.	Y	7.654	Y	19441.434	0.000	2.5400
	Vert. Up	Y	2.176	Y	33167.953	0.000	15.2400
	Vert. Dn	Y	20.633	Y	1257770.0	0.000	60.9600

SOIL PARAMETERS (24SAND)

Calculation Method : American Lifeline Alliance (ASCE 2001)
Soil Type : Dense Sand
Pipe Direction : Horizontal

Parameters	Low	High	Average
Outside Diameter, D [mm]	609.60		
Depth to Centerline, H [mm]	1524.00		
Effective Unit Wt. above pipe [kg/m3]	1601.85	1601.85	1601.85
Total Unit Wt. below pipe [kg/m3]	1601.85	1601.85	1601.85
Dry Unit Wt. above pipe [kg/m3]	1601.85	1601.85	1601.85
Soil Cohesion, c [N/m2]	0.00	0.00	0.00
Friction Angle, phi [deg]	40.00	40.00	40.00
Coating Factor, f	0.80	0.80	0.80
Coefficient of pressure at rest, K0	0.36	0.36	0.36
Horizontal Yield Displacement, dp [mm]	60.96	60.96	60.96
Longitudinal Yield Displacement, dt [mm]	2.54	2.54	2.54
Vertical Up Yield Displacement, dqu [mm]	15.24	15.24	15.24
Vertical Dn Yield Displacement, dqd [mm]	60.96	60.96	60.96
*Height of water on top of pipe [mm]	0.00	0.00	0.00
*Soil Adhesion, Ca [N/m2]	47880.00	47880.00	47880.00

Note: Fields with * are used in soil overburden calculations only and do not effect soil stiffness calculations.

Computed soil parameters (ASCE Method):

		Low	High	Average
Longitudinal	Adhesion alpha	0.00	0.00	0.00
	Pipe/Soil delta=f*phi [deg]	32.00	32.00	32.00
Horizontal	Nch	0.00	0.00	0.00
	Nqh	15.61	15.61	15.61
Vertical Up	Ncv	0.00	0.00	0.00
	Nqv	2.27	2.27	2.27
	Soil Weight on top Ws [N/m]	14593.90	14593.90	14593.90
Vertical Down	Nc	75.32	75.32	75.32
	Nq	64.20	64.20	64.20
	Ngamma	109.95	109.95	109.95

Derived values for buried piping :

Circumferential wall bending stress method: Adams et. al.

	Low	High	Average
Circumferential wall bending stress [N/mm2]	19.58	19.58	19.58
Linear buoyancy force on pipe, Fb [N/m]	0.00	0.00	0.00
Equiv. seismic wave prop. temperature [deg C]	-17.78	-17.78	-17.78
Buoyancy reduction factor R	1.00	1.00	1.00

Burial factor B' (Adams method - for Pipe ID 24"CL900) 0.22
Burial factor B' (CC N-755-1 - for Pipe ID 24"CL900) 0.24

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                                VIRTUAL ANCHOR DATA (24SAND)
                                -----------------------------

  Pipe Identifier            :   24"CL900
  Point name for temp. data  :   A00
  Operating Case             :   OP1
  Thermal Expansion          :       0.415 mm/m
  Temperature Change         :      36.000 deg C
  Operating Pressure         :      11.000 N/mm2
  Virtual Anchor lengths  Lm :   186426.22 mm
                          La :   372852.44 mm
                          Lb :    9644.613 mm

                                EARTH LOADS (24SAND)
                                ------------------------

  Pipe Identifier                                :  24"CL900
  Surface live load file name                    :
  Pressure on pipe due to surface load       Pp :       0.000 N/mm2
  Trench laying condition                        :  Type 1
  Modulus of passive soil fill reaction      E' :       1.034 N/mm2
  Bending moment coefficient                 Kb :       0.235
  Deflection coefficient                     Kx :       0.108
  Deflection lag factor                      Dl :       1.500
  Bedding constant                           K  :       0.100
  Soil support factor                        Fs :       0.100

                                SEISMIC LOADS (24SAND)
                                ------------------------

  Pipe Identifier                                :  24"CL900
  Soil input parameters set                      :  Low
  Seismic wave type                              :  Compression
  Seismic wave coefficient                   aw :       1.000
  Seismic wave curvature coefficient         ak :       1.600
  Wave length of seismic wave                Lw :   2540.000 mm
  Seismic wave velocity at pipeline          c  :  1066800.0 mm/s
  Peak ground acceleration                  PGA :   2540.000 mm/s2
  Peak ground velocity                      PGV :    381.000 mm/s
  Add axial and bending strain                   :  N

                                UPHEAVAL BUCKLING (24SAND)
                                -------------------------

  Consider Upheaval Buckling                     :  N
  Use K. Peters Compressive Force Calculation :  N
```

L O A D S   S U M M A R Y   D A T A   L I S T I N G

WIND LOAD CASES :

Number of load cases : 4

Ground elevation for wind : 2000.00 mm

Wind shape factor multiplier for piping: 1.000

Wind shape factor multiplier for beams: 1.000

Exposure factor for buried pipes : 0.000

Exposure factor for submerged pipes : 0.000

Exposed Segments - B C G I O P W AE AZ BA

Wind application method : Projected

L O A D S   S U M M A R Y   D A T A   L I S T I N G

Load case 1 - W1

Direction - Global X-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient
Cf = Force coefficient or wind shape factor
Qz = Velocity pressure = 0.00256*Kz*Kzt*V^2 (Eq. 27.3-1)
Pressure = Qz*G*Cf (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"
Default Gust Factor G assumes pipe frequency > 1 Hz.
Piping is assumed not to be on an isolated hill (Kzt=1).
Automatic Cf reported in chart above is calculated
assuming D=1 ft. Program uses actual Pipe OD to
calculate Cf at height during analysis.

L O A D S   S U M M A R Y   D A T A   L I S T I N G

Load case 1 - W1

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient
Cf = Force coefficient or wind shape factor
Qz = Velocity pressure = 0.00256*Kz*Kzt*V^2 (Eq. 27.3-1)
Pressure = Qz*G*Cf (Eq. 23)

L O A D S   S U M M A R Y   D A T A   L I S T I N G

Load case 2 - W2

Direction - Global -X-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient
Cf = Force coefficient or wind shape factor
Qz = Velocity pressure = 0.00256*Kz*Kzt*V^2 (Eq. 27.3-1)
Pressure = Qz*G*Cf (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"
Default Gust Factor G assumes pipe frequency > 1 Hz.
Piping is assumed not to be on an isolated hill (Kzt=1).
Automatic Cf reported in chart above is calculated
assuming D=1 ft. Program uses actual Pipe OD to
calculate Cf at height during analysis.

L O A D S   S U M M A R Y   D A T A   L I S T I N G

Load case 2 - W2

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient
Cf = Force coefficient or wind shape factor
Qz = Velocity pressure = 0.00256*Kz*Kzt*V^2 (Eq. 27.3-1)
Pressure = Qz*G*Cf (Eq. 23)

L O A D S   S U M M A R Y   D A T A   L I S T I N G

Load case 3 - W3

Direction - Global Z-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient
Cf = Force coefficient or wind shape factor
Qz = Velocity pressure = 0.00256*Kz*Kzt*V^2 (Eq. 27.3-1)
Pressure = Qz*G*Cf (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"
Default Gust Factor G assumes pipe frequency > 1 Hz.
Piping is assumed not to be on an isolated hill (Kzt=1).
Automatic Cf reported in chart above is calculated
assuming D=1 ft. Program uses actual Pipe OD to
calculate Cf at height during analysis.

L O A D S   S U M M A R Y   D A T A   L I S T I N G

Load case 3 - W3

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient
Cf = Force coefficient or wind shape factor
Qz = Velocity pressure = 0.00256*Kz*Kzt*V^2 (Eq. 27.3-1)
Pressure = Qz*G*Cf (Eq. 23)

L O A D S   S U M M A R Y   D A T A   L I S T I N G

Load case 4 - W4

Direction - Global -Z-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient
Cf = Force coefficient or wind shape factor
Qz = Velocity pressure = 0.00256*Kz*Kzt*V^2 (Eq. 27.3-1)
Pressure = Qz*G*Cf (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"
Default Gust Factor G assumes pipe frequency > 1 Hz.
Piping is assumed not to be on an isolated hill (Kzt=1).
Automatic Cf reported in chart above is calculated
assuming D=1 ft. Program uses actual Pipe OD to
calculate Cf at height during analysis.

L O A D S   S U M M A R Y   D A T A   L I S T I N G

Load case 4 - W4

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient
Cf = Force coefficient or wind shape factor
Qz = Velocity pressure = 0.00256*Kz*Kzt*V^2 (Eq. 27.3-1)
Pressure = Qz*G*Cf (Eq. 23)

S U P P O R T   D A T A   L I S T I N G

```
 Point Support  Support Conn.to Comp.Wt  Stiff.     Gap 1   Gap 2  Fric. GapSet Preload Ld.Var Size    Figure
 Name   Type      ID     /Dir    (kg)               (mm)    (mm)   Fact. /#hgr   (N  )
 ----- -------- ------- ------- ------- ---------- ------- ------- ----- ------ ------- ------ ---- ------------

 Tag No.: <None>                                Attachment ID: <None>
 B01   Shoe     B01  1  Ground    0.000 Rigid          0.0  2540.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 B08   Guide-V  B08  1  Ground    0.000 Rigid          0.0     3.0  0.10  Weightless
       Guide-H  B08  1  Ground    0.000 Rigid          3.0     3.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 O01   Guide-V  O01  1  Ground    0.000 Rigid          0.0     3.0  0.10  Weightless
       Guide-H  O01  1  Ground    0.000 Rigid          3.0     3.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 O06   Shoe     O06  1  Ground    0.000 Rigid          0.0  2540.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 P01 M Shoe     P01 M1  Ground    0.000 Rigid          0.0  2540.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 P03   Guide-V  P03  1  Ground    0.000 Rigid          0.0     3.0  0.10  Weightless
       Guide-H  P03  1  Ground    0.000 Rigid          3.0     3.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 P04 M Shoe     P04 M1  Ground    0.000 Rigid          0.0  2540.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 P07 M Shoe     P07 M1  Ground    0.000 Rigid          0.0  2540.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 P10 M Shoe     P10 M1  Ground    0.000 Rigid          0.0  2540.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 W02 M Shoe     W02 M1  Ground    0.000 Rigid          0.0  2540.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 AZ04M Shoe     AZ04M1  Ground    0.000 Rigid          0.0  2540.0  0.00  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 AZ07  Guide-V  AZ07 1  Ground    0.000 Rigid          0.0     3.0  0.10  Weightless
       Guide-H  AZ07 1  Ground    0.000 Rigid          3.0     3.0  0.10  Weightless

 Tag No.: <None>                                Attachment ID: <None>
 AZ08M Shoe     AZ08M1  Ground    0.000 Rigid          0.0  2540.0  0.00  Weightless

 Spring Manufacturer: Anvil/Grinnell
 NOTE 1: Soil supports are present but not listed.
 Gap 1 : V-stop,Guide-V=down, Linestop,Incline,Tie/link=backward, Guide-H=Left
 Gap 2 : V-stop,Guide-V=Up  , Linestop,Incline,Tie/link=forward , Guide-H=Right
 Stiffness units for rotation support: N.m/deg  , all others: N/mm
```

B E N D D A T A L I S T I N G

Point Name	Bend Type	Radius (mm)	Angle (deg)	OD (mm)	tnom (mm)	Material	Flan^ /Cuts	SIF in	SIF out	Analysis Set	Flexibility (K)	Pressure Case
A01	Elbow	914L	29.2	609.6	15.9	5LX-X70	0/0	2.99	2.50	1	10.01	None
								2.99	2.50	2	10.01	None
P14	Elbow	914L	29.5	609.6	15.9	5LX-X70	0/0	2.99	2.50	1	10.01	None
								2.99	2.50	2	10.01	None
W01	Elbow	914L	90.0	609.6	15.9	5LX-X70	0/0	2.99	2.50	1	10.01	None
								2.99	2.50	2	10.01	None
W05	Elbow	914L	90.0	609.6	15.9	5LX-X70	0/0	2.99	2.50	1	10.01	None
								2.99	2.50	2	10.01	None
AZ02	Elbow	914L	29.2	609.6	15.9	5LX-X70	0/0	2.99	2.50	1	10.01	None
								2.99	2.50	2	10.01	None
BA01	Elbow	914L	29.5	609.6	15.9	5LX-X70	0/0	2.99	2.50	1	10.01	None
								2.99	2.50	2	10.01	None

^ = Number of bend ends where either a flange or valve is within distance
(L/D) x Nominal diameter, where L/D is defined under Tools > Model options > Edit
L = Long radius

T E E   D A T A   L I S T I N G

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	**Crotch Radius (mm)	*Crotch Thick (mm)	SIF in	SIF out
A02	A-Header	609.6	15.9	5LX-X70	Other			1.00	1.00
	AZ-Branch	609.6	15.9	5LX-X70					
B02	B-Header	609.6	15.9	5LX-X70	Other			1.00	1.00
	G-Branch	114.3	17.1	5LS-B					
B05	B-Header	711.2	24.0	5LX-X70	Other			1.00	1.00
	C-Branch	219.1	18.3	5L-B					
B07	B-Header	711.2	24.0	5LX-X70	Other			1.00	1.00
	I-Branch	114.3	17.1	5LS-B					
O03	O-Header	711.2	24.0	5LX-X70	Welding			1.73	1.98
	AE-Branch	114.3	17.1	5LS-B					
P06	P-Header	609.6	15.9	5LX-X70	Welding			2.02	2.36
	W-Branch	609.6	15.9	5LX-X70					
P12	P-Header	609.6	15.9	5LX-X70	Welding			2.02	2.36
	W-Branch	609.6	15.9	5LX-X70					

* = Pad thickness for reinforced tee.
**= Fillet radius for Raised tee or discontinuity dist. for Thickened tee.

V A L V E   D A T A   L I S T I N G

Point Name	Type	Dnom (mm)	Length (mm)	Valve Weight (kg)	Auto	Symbol	Actuator Weight (kg)	DX (mm)	DY (mm)	DZ (mm)	SAF	Joint Type	SIF	Size /Avg. (mm)	Type /Max. (mm)
Tag No. : MV1017															
P01	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1017															
P04	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1017															
P07	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1017															
P10	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1017															
W02	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : SDV-1003															
AZ04	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1057															
AZ08	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		

C A T A L O G   V A L V E   D A T A   L I S T I N G

```
 Point Name  : P01
 Manufacturer: AutoPIPE Generic
 Standard    : ANSI/ASME
 Sub Category: Ball Valves
 Type        : Ball Valve Side Entry
 Main End    : FL
 Rating      : 900

 Point Name  : P04
 Manufacturer: AutoPIPE Generic
 Standard    : ANSI/ASME
 Sub Category: Ball Valves
 Type        : Ball Valve Side Entry
 Main End    : FL
 Rating      : 900

 Point Name  : P07
 Manufacturer: AutoPIPE Generic
 Standard    : ANSI/ASME
 Sub Category: Ball Valves
 Type        : Ball Valve Side Entry
 Main End    : FL
 Rating      : 900

 Point Name  : P10
 Manufacturer: AutoPIPE Generic
 Standard    : ANSI/ASME
 Sub Category: Ball Valves
 Type        : Ball Valve Side Entry
 Main End    : FL
 Rating      : 900

 Point Name  : W02
 Manufacturer: AutoPIPE Generic
 Standard    : ANSI/ASME
 Sub Category: Ball Valves
 Type        : Ball Valve Side Entry
 Main End    : FL
 Rating      : 900

 Point Name  : AZ04
 Manufacturer: AutoPIPE Generic
 Standard    : ANSI/ASME
 Sub Category: Ball Valves
 Type        : Ball Valve Side Entry
 Main End    : FL
 Rating      : 900

 Point Name  : AZ08
 Manufacturer: AutoPIPE Generic
 Standard    : ANSI/ASME
 Sub Category: Ball Valves
 Type        : Ball Valve Side Entry
 Main End    : FL
 Rating      : 900
```

F L A N G E   D A T A   L I S T I N G

Point Name	FLG No.	Type	Dnom (mm)	Rating	Flange Weight (kg)	Bolt Weight (kg)	Joint Type	SIF	Size /Avg. (mm)	Type /Max. (mm)	ANSI Check
Tag No. : <None>											
B09	1	BLIND	700	900	1400.00	700.00	USER	1.00			No
Tag No. : <None>											
C01	1	WELDNECK	200	900	84.82	0.00	WN	1.00			No
Tag No. : <None>											
G01	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. : <None>											
G01	2	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. : <None>											
I01	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. : <None>											
O00	1	BLIND	700	900	1400.00	700.00	USER	1.00			No
Tag No. : <None>											
P01	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
P02	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
P04	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
P05	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
P07	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
P08	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
P10	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
P11	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
W02	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
W03	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. : <None>											
AE01	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. : <None>											
AZ04	1	WELDNECK	600	900	1764.00	882.00	WN	1.00			No
Tag No. : <None>											
AZ05	1	WELDNECK	600	900	1764.00	882.00	WN	1.00			No
Tag No. : <None>											
AZ08	1	WELDNECK	600	900	1764.00	882.00	WN	1.00			No
Tag No. : <None>											
AZ09	1	WELDNECK	600	900	1764.00	882.00	WN	1.00			No

R E D U C E R   D A T A   L I S T I N G

From Point	To Point	Length (mm)	OD From (mm)	OD To (mm)	Thk From (mm)	Thk To (mm)	From Material	Cone Angle (deg)	SIF
B03	B04	620.00	609.6	711.2	15.9	24.0	5LX-X70		2.00t
O04	O05	620.00	711.2	609.6	24.0	15.9	5LX-X70		2.00t

t = Use SIF of 2.0

C E N T E R   O F   G R A V I T Y   R E P O R T

Description		Weight (kg)	X CoG (mm)	Y CoG (mm)	Z CoG (mm)
Valve	:	47500.99	29929.00	7500.00	-2400.00
Support	:	0.00	0.00	0.00	0.00
Flange	:	32601.34	32734.56	7503.81	-2010.15
Flexible Joint	:	0.00	0.00	0.00	0.00
Additional Weight	:	7190.00	20511.10	7500.00	-1475.00
Pipes (Run, Bend, Tee, Reducer)	:	198913.00	-155330.8	2423.79	-1488.54
PipeTot (Valve+Support+Flange+Flex. Jt.+Add. Wt.+Pipes)	:	286205.25	-98743.79	3972.47	-1698.89
Insulation	:	0.00	0.00	0.00	0.00
Cladding	:	0.00	0.00	0.00	0.00
Lining	:	0.00	0.00	0.00	0.00
Contents	:	158843.48	-153688.9	2472.54	-1506.00
Beam	:	0.00	0.00	0.00	0.00
PipeTot + Ins.	:	286205.25	-98743.79	3972.47	-1698.89
PipeTot + Ins. + Clad.	:	286205.25	-98743.79	3972.47	-1698.89
PipeTot + Ins. + Clad. + Lin.	:	286205.25	-98743.79	3972.47	-1698.89
PipeTot + Ins. + Clad. + Lin. + Cont.	:	445048.84	-118354.4	3437.12	-1630.04
PipeTot + Beam	:	286205.25	-98743.79	3972.47	-1698.89
PipeTot + Ins. + Beam	:	286205.25	-98743.79	3972.47	-1698.89
PipeTot + Ins. + Clad. + Beam	:	286205.25	-98743.79	3972.47	-1698.89
PipeTot + Ins. + Clad. + Lin. + Beam	:	286205.25	-98743.79	3972.47	-1698.89
PipeTot + Ins. + Clad. + Lin. + Cont. + Beam	:	445048.84	-118354.4	3437.12	-1630.04

NOTE: Fluid Density Factor is assumed as 1.0.
NOTE: COG report only includes the currently visible beams and segments.
All pipe segments are included.
No beams present in the model.

L O A D   C A S E   D E S C R I P T I O N

Load Case	Load Case Name	Description
GR	Gravity	
T1	Thermal 1	
T2	Thermal 2	
T3	Thermal 3	
W1	Wind 1	
W2	Wind 2	
W3	Wind 3	
W4	Wind 4	
P1	Press 1	
P2	Press 2	
P3	Press 3	

HYDROTEST LOAD SUMMARY

```
 Specific Gravity of Fluid    :    1.000

 Hydrotest Pressure Case      : P3
 Hydrotest Pressure Factor    :    1.500

 Hydrotest Temperature Case   : T3
 Hydrotest Temperature Factor :    1.000

 Cut-short Case               : None
 Include Insulation & Cladding: No
```

ADDITIONAL WEIGHT DATA LISTING

Point Name	Added Weight kg	DX mm	DY mm	DZ mm
P13	3595.00			
AZ03	3595.00			

Point Summary

Number of Points in the System					Weight of System(kg)		
Pipe	Frame	Soil	Total	Mass	Pipe	Contents	Total
91	0	1527	1618	0	286205.3	158843.5	445048.8

ANCHOR DATA LISTING

	Translational Stiffness			Rotational Stiffness			Release Hanger		
Point Name	KTX N/mm	KTY N/mm	KTZ N/mm	KRX N.m/deg	KRY N.m/deg	KRZ N.m/deg	X XX	Y YY	Z ZZ
A00				Connected to: Ground					
	RIGID	RIGID	RIGID	RIGID	RIGID	RIGID			
P09				Connected to: Ground					
	RIGID	RIGID	RIGID	RIGID	RIGID	RIGID			
AZ06				Connected to: Ground					
	RIGID	RIGID	RIGID	RIGID	RIGID	RIGID			
BA02				Connected to: Ground					
	RIGID	RIGID	RIGID	RIGID	RIGID	RIGID			

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	Offset (mm) DX	DY	DZ	Global (mm) X	Y	Z
*** Segment A										
Origin	A00	Point	24"CL900	0.00	-369638.00	1900.00	0.00	-369638.00	1900.00	0.00
A00	A01	Bend	24"CL900	31517	378203.03	0.00	0.00	8565.00	1900.00	0.00
A01	A02	Tee	24"CL900	477.55	5000.00	2800.00	0.00	13565.00	4700.00	0.00
*** Segment B										
Origin	B00	Point	24"CL900	0.00	36931.20	7500.00	0.00	36931.20	7500.00	0.00
B00	B01	Junction	24"CL900	108.33	1300.00	0.00	0.00	38231.20	7500.00	0.00
B01	B02	Tee	24"CL900	116.67	1400.00	0.00	0.00	39631.20	7500.00	0.00
B02	B03	Run	24"CL900	208.33	2500.00	0.00	0.00	42131.20	7500.00	0.00
B03	B04	Reducer	28"PIPE	51.67	620.00	0.00	0.00	42751.20	7500.00	0.00
B04	B05	Tee	28"PIPE	50.00	600.00	0.00	0.00	43351.20	7500.00	0.00
B05	B06	Run	28"PIPE	120.83	1450.00	0.00	0.00	44801.20	7500.00	0.00
B06	B07	Tee	28"PIPE	245.83	2950.00	0.00	0.00	47751.20	7500.00	0.00
B07	B08	Run	28"PIPE	141.67	1700.00	0.00	0.00	49451.20	7500.00	0.00
B08	B09	Run	28"PIPE	100.00	1200.00	0.00	0.00	50651.20	7500.00	0.00
*** Segment C										
Origin	B05	Tee	8"C03	0.00	43351.20	7500.00	0.00	43351.20	7500.00	0.00
B05	C01	Run	8"C03	58.33	0.00	700.00	0.00	43351.20	8200.00	0.00
*** Segment G										
Origin	B02	Tee	4"C06	0.00	39631.20	7500.00	0.00	39631.20	7500.00	0.00
B02	G01	Run	4"C06	58.33	0.00	700.00	0.00	39631.20	8200.00	0.00
*** Segment I										
Origin	B07	Tee	4"C06	0.00	47751.20	7500.00	0.00	47751.20	7500.00	0.00
B07	I01	Run	4"C06	58.33	0.00	700.00	0.00	47751.20	8200.00	0.00
*** Segment O										
Origin	O00	Point	28"PIPE	0.00	50651.20	7500.00	-2950.00	50651.20	7500.00	-2950.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	Offset (mm) DX	DY	DZ	Global (mm) X	Y	Z
O00	O01	Run	28"PIPE	113.33	-1360.00	0.00	0.00	49291.20	7500.00	-2950.00
O01	O02	Run	28"PIPE	75.00	-900.00	0.00	0.00	48391.20	7500.00	-2950.00
O02	O03	Tee	28"PIPE	350.00	-4200.00	0.00	0.00	44191.20	7500.00	-2950.00
O03	O04	Run	28"PIPE	125.00	-1500.00	0.00	0.00	42691.20	7500.00	-2950.00
O04	O05	Reducer	24"CL900	51.67	-620.00	0.00	0.00	42071.20	7500.00	-2950.00
O05	O06	Run	24"CL900	62.50	-750.00	0.00	0.00	41321.20	7500.00	-2950.00
O06	O07	Run	24"CL900	62.50	-750.00	0.00	0.00	40571.20	7500.00	-2950.00
*** Segment P										
Origin	O07	Point	24"CL900	0.00	40571.20	7500.00	-2950.00	40571.20	7500.00	-2950.00
O07	P01	Junction	24"CL900	1.67	-20.00	0.00	0.00	40551.20	7500.00	-2950.00
P01	P02	Valve	24"CL900	190.50	-2286.00	0.00	0.00	38265.20	7500.00	-2950.00
P02	P03	Run	24"CL900	170.83	-2050.00	0.00	0.00	36215.20	7500.00	-2950.00
P03	P04	Run	24"CL900	225.00	-2700.00	0.00	0.00	33515.20	7500.00	-2950.00
P04	P05	Valve	24"CL900	190.50	-2286.00	0.00	0.00	31229.20	7500.00	-2950.00
P05	P06	Tee	24"CL900	158.33	-1900.00	0.00	0.00	29329.20	7500.00	-2950.00
P06	P07	Run	24"CL900	83.33	-1000.00	0.00	0.00	28329.20	7500.00	-2950.00
P07	P08	Valve	24"CL900	190.50	-2286.00	0.00	0.00	26043.20	7500.00	-2950.00
P08	P09	Run	24"CL900	79.17	-950.00	0.00	0.00	25093.20	7500.00	-2950.00
P09	P10	Run	24"CL900	79.17	-950.00	0.00	0.00	24143.20	7500.00	-2950.00
P10	P11	Valve	24"CL900	190.50	-2286.00	0.00	0.00	21857.20	7500.00	-2950.00
P11	P12	Tee	24"CL900	83.33	-1000.00	0.00	0.00	20857.20	7500.00	-2950.00
P12	P13	Run	24"CL900	116.67	-1400.00	0.00	0.00	19457.20	7500.00	-2950.00
P13	P14	Bend	24"CL900	250.00	-3000.00	0.00	0.00	16457.20	7500.00	-2950.00
P14	P15	Run	24"CL900	483.64	-5050.00	-2860.00	0.00	11407.20	4640.00	-2950.00
*** Segment W										
Origin	P06	Tee	24"CL900	0.00	29329.20	7500.00	-2950.00	29329.20	7500.00	-2950.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	Offset (mm) DX	DY	DZ	Global (mm) X	Y	Z
P06	W01	Bend	24"CL900	170.83	0.00	0.00	-2050.00	29329.20	7500.00	-5000.00
W01	W02	Run	24"CL900	141.67	-1700.00	0.00	0.00	27629.20	7500.00	-5000.00
W02	W03	Valve	24"CL900	190.50	-2286.00	0.00	0.00	25343.20	7500.00	-5000.00
W03	W04	Run	24"CL900	186.92	-2243.00	0.00	0.00	23100.20	7500.00	-5000.00
W04	W05	Bend	24"CL900	186.92	-2243.00	0.00	0.00	20857.20	7500.00	-5000.00
W05	P12	Tee	24"CL900	170.83	0.00	0.00	2050.00	20857.20	7500.00	-2950.00
*** Segment AE										
Origin	O03	Tee	4"C06	0.00	44191.20	7500.00	-2950.00	44191.20	7500.00	-2950.00
O03	AE01	Run	4"C06	58.33	0.00	700.00	0.00	44191.20	8200.00	-2950.00
*** Segment AZ										
Origin	AZ00	Point	24"CL900	0.00	8565.00	1900.00	0.00	8565.00	1900.00	0.00
AZ00	A02	Tee	24"CL900	477.55	5000.00	2800.00	0.00	13565.00	4700.00	0.00
A02	AZ02	Bend	24"CL900	477.55	5000.00	2800.00	0.00	18565.00	7500.00	0.00
AZ02	AZ03	Run	24"CL900	250.00	3000.00	0.00	0.00	21565.00	7500.00	0.00
AZ03	AZ04	Run	24"CL900	200.05	2400.60	0.00	0.00	23965.60	7500.00	0.00
AZ04	AZ05	Valve	24"CL900	190.50	2286.00	0.00	0.00	26251.60	7500.00	0.00
AZ05	AZ06	Run	24"CL900	41.67	500.00	0.00	0.00	26751.60	7500.00	0.00
AZ06	AZ07	Run	24"CL900	362.23	4346.80	0.00	0.00	31098.40	7500.00	0.00
AZ07	AZ08	Run	24"CL900	308.33	3700.00	0.00	0.00	34798.40	7500.00	0.00
AZ08	AZ09	Valve	24"CL900	190.50	2286.00	0.00	0.00	37084.40	7500.00	0.00
AZ09	B00	Run	24"CL900	12.77	-153.20	0.00	0.00	36931.20	7500.00	0.00
*** Segment BA										
Origin	P15	Point	24"CL900	0.00	11407.20	4640.00	-2950.00	11407.20	4640.00	-2950.00
P15	BA01	Bend	24"CL900	483.64	-5050.00	-2860.00	0.00	6357.20	1780.00	-2950.00
BA01	BA02	Run	24"CL900	31250	-375000.00	0.00	-0.02	-368642.78	1780.00	-2950.02